Supporting Information for "Hidden support share as a conditional risk screen for annual thermal extremes in incomplete river-temperature records"

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Introduction

This supporting information records what belongs to the record rather than to the argument: data provenance, protocol history, the predictions declared for this round that were not met, the runs that were discarded and why, and the reproduction path. Nothing here is required to follow the article.

S1 Data sources

Daily water-temperature series were assembled from nine public providers (Table S1); two further agencies (SMHI, UK Environment Agency) supply the never-developed transport panel. Each provider's product, documented statistic, quality codes, licence and acquisition route were re-verified against the provider's own documentation on 2026-10-12 (`results/wrr_submission_ready/measurement_provenance_dictionary.csv`). Eight of the eleven publish under a named open licence and none required a data-use agreement, but LUBW reserves copyright on its raw values: those seven networks are not redistributed here and are read locally through LUBW's own service.

Station coordinates were assembled for all 457 stations: from the providers' own catalogues already held in the repository for FOEN, CHMI, SYKE, LUBW, GKD Bayern, NVE and Rijkswaterstaat and for nine USGS gauges; from the USGS

Table S1. Providers and their role in the study.

provider	country	role
ARSO	Slovenia	development and reserved
CHMI	Czechia	development and reserved
FOEN	Switzerland	development
GKD Bayern	Germany (Bavaria)	development and reserved
LUBW	Germany (Baden-Württemberg)	development and reserved
NVE	Norway	development and reserved
Rijkswaterstaat	Netherlands	development
SYKE	Finland	development
USGS	United States	development and reserved

NWIS site service for the remaining 195 United States gauges; and from ARSO's public station feed for all 47 Slovenian stations. Every row of the coordinate table in the repository records its source.

Daily 2 m air temperature is NASA POWER (MERRA-2), 1 January 1981 to 31 August 2026, one point per station at the product's native resolution: 16,679 days for each of the 457 stations. An earlier version of this study used a much smaller forcing table covering 2.3% of development cases; both are retained and Section 4.4.2 of the article compares them.

Basin characteristics for the United States subset are GAGES-II, joined on the eight-digit gauge identifier and matching 140 of 204 eligible stations.

S2 Quality control and the definition of a network

A *network* is a provider's group of stations on a common river system, which is the unit at which donor stations are available to one another. Networks are the clustering unit for every interval in the article, because station-years within a network share weather, regulation and instrumentation.

Waterbody class was assigned per network from the provider's own naming and an explicit map; seas, lakes, canals, estuaries and deltas are listed and excluded

rather than averaged into a river result. Reference series are never imputed: a calendar year with any missing day contributes no truth. Stations with fewer than four fully observed calendar years are excluded, because the leave-one-year-out estimator protocol requires them. The resulting denominator, with every exclusion and its count, is in the repository. The same stations contribute 2422 station-years with at least 360 daily slots and at least one missing day; those form the deployment panel of Section 4.4.3 of the article.

Three measurement strata are reported rather than merged. In the study panel, five providers document a daily mean that the parser reads directly (USGS, FOEN, GKD Bayern, NVE, LUBW); two are project-derived daily means of the provider's own sub-daily values (CHMI's T_O instantaneous series, Rijkswaterstaat's raw `N V T` readings); and two are not a documented daily mean at all (SYKE's 08:00 nearshore reference reading, ARSO's undocumented statistic). The transport panel adds the UK Environment Agency, which documents a daily mean, and SMHI and Hub'Eau, whose daily statistics are undocumented and continuous respectively; those sit in the assurance strata recorded by the transport seal. Restricting the development panel to the documented-daily-mean stratum gives a contrast of -5.7 points ($[-6.5, -4.9]$, 84 networks) against -6.4 for the full panel, and the reserved panel -6.6 ($[-7.9, -5.3]$, 29 networks) against -7.1; the sealed confirmation is USGS-only and lies entirely inside the documented stratum (`amendments/provider_contract_sensitivity/`). Each provider's product, documented statistic, quality codes, licence and acquisition route are tabulated with verbatim evidence in `measurement_provenance_dictionary.csv` and `current_data_acquisition.csv`.

Transport-panel panels receive one additional structural quality-control pass before sealing, applied identically to every provider and recorded before any score: warm-season (May–September) runs of three or more consecutive days at exactly 0.0 °C are blanked as stuck-sensor values. Two such runs were found in the UK Environment Agency ingest, one of them inside a hottest-30 window; the rule and its counts are in `multi_agency_ingest/QC_SUMMARY.json` and the per-provider copies in `multi_agency_ingest/uk_ea/qc_report.csv` and `multi_agency_ingest/smhi/qc_report.csv`.

S3 Protocol history

The study was pre-registered before the reserved networks were opened on 2026-09-23. Three aspects of that freeze were incomplete or wrong and are recorded here rather than silently repaired.

The incumbent's imputed limit was mis-stated. An earlier version placed the comparison at a thirteen-week (91-day) limit attributed to Johnson et al. (2021). The source says twenty weeks. The analysis code now uses 140 days and every artifact that used the old value was re-run; Section 4.4.1 of the article reports the corrected operating point. The direction of the error flattered the incumbent at that point, so correcting it reduces the screen's advantage there from 6.2 to 4.4 percentage points.

The weighting between outage arms was not declared. The case set mixes an empirical arm, drawn from the outage geometry the same networks experience, with a stratified arm built to span the decision space. Their base rates differ by an order of magnitude. The pre-registered test was run on the pooled set, as declared; Section 4.2.3 reports the decomposition and states that the declared effect is not met on the empirical arm alone.

The natural scale of the peak has two values. Computed over the full eligible roster it is 1.06 °C; recomputed inside the development case table it is 1.08 °C. The primary tolerance is 0.50 °C throughout except in cross-metric comparisons, where each endpoint's scale is recomputed on the same footing. The tolerance sweep shows the axis ordering is unchanged between 0.25 and 1.0 standard deviations.

S4 Confirmation, transport and the amendment register

The first sealed confirmation (executed; audit-failed; withdrawn). Thirty-nine USGS HUC4 networks outside the v17/v18 development panel were ingested, hashed and scored once. The historical pooled contrast of about −7.9 points is retained only as an executed artifact. The 2026-10-13 audit finds USGS no-data sentinels in truth years and silent deletion of non-reconstructable selected cases before the 70% cut (`AMENDMENT_claim_status_freeze_20261013.md`; `amendments/p0_02_qc_ledger/`; `amendments/p0_03_retention_denominator/`). That seal is

withdrawn. Prior-access diagnostics in `AMENDMENT_prior_access_sealed_panel.md` remain relevant for any future reanalysis of this opened panel.

Wave-2 independent confirmation (scored once; success rule met). Under `P0_07_NEW_CONFIRMATION_SEAL.json`, 49 networks / 143 stations / 13 017 cases across USGS, Hub'Eau and UK EA were assembled with programme-used stations excluded, hashed, and scored once with repaired QA and reconstructability. At 70% retention and 0.5 °C the contrast is $\Delta = -5.89$ points ($[-6.92, -4.91]$), with zero reconstruction failures, USGS and Hub'Eau both negative, and the USGS quality-checked subset agreeing (`AMENDMENT_p0_07_wave2_confirmation_20261013.md`; `p0_07_wave2_ingest/p0_07_wave2_confirmation/verdict.json`).

The transport test. The confirmation is single-agency, so the pre-registered transport protocol (`MULTI_AGENCY_TRANSPORT_PROTOCOL.md`) recruits agencies absent from the v17 roster and scores the same frozen configuration once on them. Three amendments were recorded before sealing: prior access is judged per station against every artifact in the repository (metadata catalogues keyed by `site_id` carry no series values and are not series use), providers whose published values are labelled uncontrolled form a separate assurance stratum reported alongside the pooled verdict, and networks from a provider that took part in development may enter only as a labelled secondary stratum. The protocol fixes the primary endpoint, tolerance, retention, the pooled decision rule ($\Delta \leq -1.0$ percentage points with a clustered interval excluding zero and at least two providers in the same direction) and the requirement that the pooled and quality-checked-only verdicts agree. It also discloses that it was designed after the USGS score was known, so it is a transport test and never a pristine first confirmation.

The sealed transport panel holds 53 networks: 32 primary networks from Sweden (SMHI, 16), England (UK Environment Agency, 2) and France (Hub'Eau, 14) across three climate contexts, plus 21 Norwegian (NVE) basins as the secondary stratum only (152 primary stations; 245 including secondary). Historical contrasts that reported a primary pooled −4.6 points and a quality-checked −6.2 points are executed artifacts only. The 2026-10-13 audit finds SMHI unit/coding failures, reconstructability dropouts that shrank the 70% denominator, and a quality-checked gate that mixed NVE secondary cases ($\approx$ 97% of that stratum). On the primary

quality-checked subset alone (2 networks, 307 cases) the corrected diagnostic fails the pre-registered CI rule (`amendments/f03_primary_qc_diagnostic/`). That transport seal remains withdrawn; it is not the wave-2 confirmation above.

The amendment register. Each adversarial finding closed after the freeze has an amendment directory holding the recomputation, the preserved pre-fix bytes and a report: instrument error under a locked method (`step22_noise_post_fix`), support-width uncertainty and the fixed-centre dilation test (`e6_support_width_uncertainty`), exact-donor-count mechanism control (`step5_mechanism_equal_k`), the climatological-peak proxy's misclassification rate (`peak_proxy_bridge`), provider-contract sensitivity (`provider_contract_sensitivity`), a faithful reimplementation of the cited air-regression baseline (`faithful_baseline`), hidden-bias sensitivity (`mnar_pattern_mixture`), cold-season support axes (`winter_regulatory`), and the sealed panel's prior-access diagnostic (`sealed_usgs_prior_access_diagnostic`). The maintenance counterfactual was recomputed under multi-year outage placement (`e1_maintenance_recompute`) and remains exploratory.

The major-revision round is registered under `major_revision_20261014/` and holds ten further analyses, each with its own `SUMMARY.md`, protocol fingerprint (script and input hashes plus the git head) and machine-readable tables: the converged method and claim set (`p0_01_final_method`), admission thresholds chosen on calibration networks only (`p0_02_frozen_risk_thresholds`), the final rule on the monitoring-calendar replay with explicit unavailable and fallback states (`p0_03_final_method_replay`), the hot-year decomposition (`p0_04_env_closed_loop`), the power and independence design (`p0_05_power_independence`), the coverage and measurement contract with an independent recompute of thirty headline values (`p0_06_coverage_recompute`), repeated learning curves (`p1_learning_curve_repeats`), the mechanism boundary with the air reconstructor and a design-based phase interval (`p1_mechanism_boundary`), and the transfer and measurement strata (`p1_transfer_measurement`). Two of these record a corrected number rather than a new one: the admission-threshold analysis replaces an operating point that had been chosen with the evaluation labels, and the replay replaces a donor-reconstruction column that had silently contained observed-window values.

S5 Measurement contract and coverage

Every result in the article is reported against the daily-value statistic its providers actually publish (Table S2). Of the 40704 development cases, 74.4 % come from the five providers that document a daily mean (84 networks), 15.5 % are project aggregates of sub-daily or instantaneous values (14 networks; CHMI ten-minute and Rijkswaterstaat raw readings), 8.3 % come from ten networks whose provider never states whether the published daily value is a mean (ARSO, 47 stations), and 1.7 % are the three Finnish morning reference readings, which the project already excludes from the unified daily-mean analysis. The retrospective reserved panel is 73.8 % documented daily means and 20.1 % project aggregates; the external wave-2 panel is 94.4 % documented daily means. A station-level coverage table with first and last year, complete years, statistic class, quality class and licence route for every station is in `major_revision_20261014/p0_06_coverage_recompute/final_coverage_table.csv`, and the results by measurement class are in the transfer table of Section 4.4.2 of the article.

Two further notes belong with that table. The reserved-panel headline contrast is sensitive to how ties at the retention cut are broken: across six tie-break seeds it runs from −7.16 to −6.94 points, so the point estimate is quoted with that range and the interval, not to two decimals. And the wave-2 complete-year count depends on the convention used by the ingest (51 of 164 stations under the rule as written, 162 under the strict all-days-present rule), so no single wave-2 station-year denominator is quoted.

S6 Parameters, their roles, and the archive

Every quantity that enters a result has one of four roles, and mixing them is a common source of confusion, so they are listed with the role stated (Table S3).

Software and data are archived as follows. The analysis code, the frozen configuration, the derived tables and the editable draw.io sources of the schematic figures are in the accompanying repository; the study inputs for the 146 study networks are packaged for confidential review (`dist/wrr_study_inputs_v1.tar.gz`) and a permanent archive with a DOI is deposited on acceptance. Restricted material is listed rather than shipped: LUBW raw values are read locally and not redistributed, and

Table S2. Daily-value statistic by provider, as documented by the provider, with the class assigned to each panel case.

provider	statistic documented	stations	class used in the analysis	evidence
USGS	daily mean	427	documented daily mean	NWIS daily-values service; parameter 00010
GKD Bayern	daily mean	55	documented daily mean	provider daily series documentation
LUBW	daily mean	22	documented daily mean	provider daily series documentation
NVE	daily mean	57	documented daily mean	HydAPI daily values
FOEN	daily mean	11	documented daily mean	opendata.swiss daily series
CHMI	sub-daily (10-minute)	73	project aggregate	provider sub-daily series
Rijkswaterstaat	raw measurements	5	project aggregate	WaterWebServices raw readings
Hub'Eau	continuous	9	project aggregate	Hub'Eau chronique
ARSO	daily values, statistic unstated	47	undocumented	provider page only
SYKE	08:00 near-shore reading	3	non-daily reference reading	provider documentation

aCounts are station-network rows from the coverage table; a station appearing in two networks is counted once per network. The class assignment is a deterministic parse of `measurement_provenance_dictionary.csv`, and the matched wording is stored per row in the coverage table.

Table S3. Model and decision parameters by role. "Fitted" means estimated from data at run time; "fixed" means a constant chosen before any result was seen; "calibrated" means chosen on networks disjoint from the cases it is applied to; "declared" means a study tolerance rather than a model quantity.

quantity	role	value or source
ridge penalty of the donor and air estimators	fixed	$\alpha = 1.0$, seven air lags, whole-year fit
gradient-boosted risk score	fitted	quantile 0.9, on the decision-time feature set; refit per fold
admission threshold	calibrated	largest retention in 0.10–0.95 whose calibration false-admit rate plus its one-sided clustered 95 % bound stays under the cap; 0.42–0.72 °C on the replay folds, 31–61 days of gap length for the simple rules
conformal multiplier	calibrated	scale factor turning the predicted bound into a valid interval, fitted on calibration networks
absolute admission rule	declared	admit when the predicted bound is at most 0.5 °C
consequential-error tolerance τ	declared	0.5 °C, i.e. 0.47 of each endpoint's interannual SD
retention operating point	declared	70 % for ranking comparisons; the risk cap (5 %) for admission comparisons
network folds	fixed	5 outer folds by network, 3 fold seeds (101/202/303), 4 inner folds for the risk score
bootstrap draws	fixed	1000 for threshold selection, 2000 for evaluation intervals

the two providers whose terms name no licence are used under attribution only (`metadata/data_rights.csv`). Input-file hashes for every network are in `current_data_manifest.csv`, and each major-revision analysis writes a protocol fingerprint holding the script hash, the input hashes and the git head.

S7 Predictions that failed

Three predictions were written down before the analyses that tested them and were not met. All three are in the article; they are gathered here because a reader looking for them should not have to hunt.

1. *The support-restricted advantage will decay monotonically in the support width.* It does not: it is a hump peaking at a 60-day support.
2. *The estimand's own support is the best conditioning set at every support width.* It is not: below 30 days a fixed 30-day window beats it.
3. *For the sine-fit phase, two hand-chosen transition windows will beat both peak-window overlap and outage length.* They beat peak overlap decisively and tie outage length.

Failures 1 and 2 have one cause, which we then measured: the support's location is itself uncertain, so the conditioning set has to be the support dilated by that uncertainty. Failure 3 has a related cause: a hand-drawn window is a crude approximation to an influence function that is spread over half the year.

Three further numbers changed under a corrected comparison rather than under new data, and the corrected values are the ones reported. The instrument-error check had evaluated its two arms with two different reconstruction choices; with one method on both arms the perturbation moves the reserved contrast from 6.21 to 6.22 points instead of from 7.06 to 6.67, and the apparent reduction disappears. The donor-count control had not actually held the donor count fixed (the paired cases differ by up to seven donors in a third of cases); an exact-match recomputation gives $+0.084\,^{\circ}\mathrm{C}$ (0.064 to 0.110), so the claim survives and the interval is restated. The support-width crossover is an interval, not a day: the bootstrap gives 77 to 107 days and the curve is a hump peaking near 60 days, not a monotone decay.

A fourth statement was *withdrawn* rather than falsified. An earlier version framed the standing guidance as getting an ordering backwards. On the guidance's own estimands it does not; Section 4.2.2 of the article reports that a completeness threshold is the correct axis for the sine-fit mean and within 0.02 AUC of the best for the phase.

S8 Runs that were discarded

Two defects made specific artifacts unusable and both were repaired before the reserved panel was opened: outage realizations were seeded from a process-salted hash, so the case table could not be regenerated, and worker results were concatenated in completion order, so the row order was not reproducible even though the content was. A third defect affected the calibration rather than the data: an additive, per-estimator conformal scheme under-covered, because selecting the smallest of several bounds is a winner's curse that per-estimator calibration cannot see. In all three cases the affected numbers were discarded rather than reported alongside the repaired ones, and the pipeline was re-run in full.

S9 Verification and reproduction

- The case table is byte-reproducible including row order, pinned by a test.
- The extended endpoint table regenerates the same 51,524 realizations from the same deterministic seeds, which is how its rows line up with the main case table; a test checks the alignment.
- No feature may read a day the outage hides; enforced in the feature constructor and pinned by a test.
- Every table cell and audited inline number in the article is re-derived from the artifacts by an audit script that the test suite runs.
- The schematic figures are generated from editable draw.io sources held in the repository; a test checks that each exported PDF is newer than its source.
- The claim registry binds every number allowed into the abstract, key points and conclusions to an artifact and is checked by `scripts/254_wrr_claim_binding_check.py`, which recomputes the allowed numbers and flags any barred number that appears in those regions.

- Every input the current study reads is listed with a byte count and a SHA-256 in `current_data_manifest.csv` (146 networks, 0 missing), and every ingest for the transport panel records its own hashes before the seal is written.
- Failed worker runs cannot leave a success table behind: the mechanism recomputation writes its summary last inside a hash-bound generation directory and re-verifies the generation on read.

Figure S1 lays the study out as a chain and labels each link with the research question that establishes it, including the two links that earlier versions of this work left open and this one closes.

S10 Supporting figures

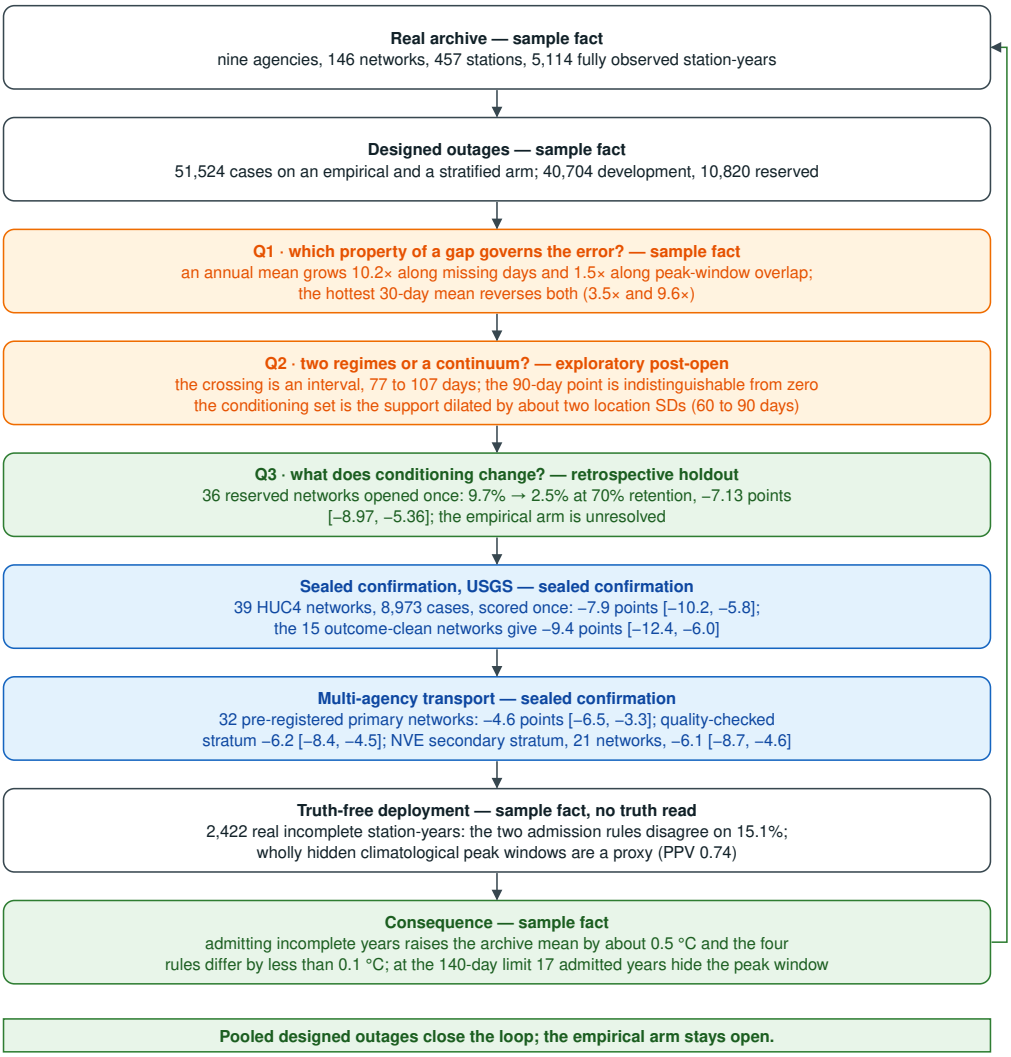


Figure S1. The evidence chain by research question. Every link is labelled with the section that establishes it and with the evidence class it rests on (sample fact, retrospective holdout, sealed confirmation, exploratory post-open); the deployment link is closed for pooled designed outages and remains open on the empirical arm, and the consequence link carries the finding that the regional aggregate barely moves.

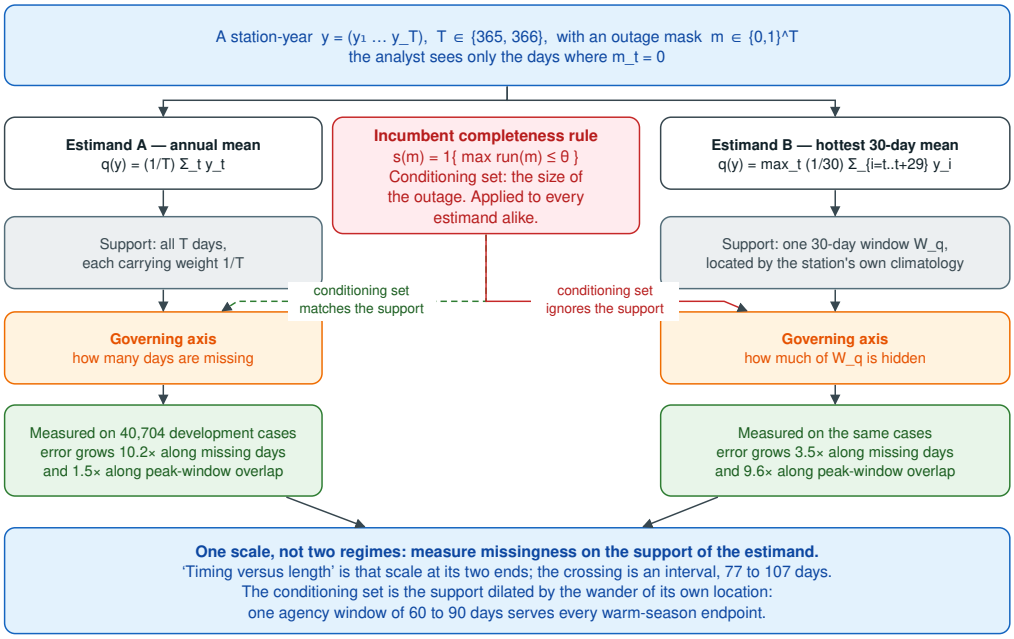


Figure S2. Problem frame (schematic), the earlier opening figure. The same station-year and outage mask feed an annual mean and a hottest 30-day mean; different supports imply different governing axes. Boxes that quote numerical gradients are mechanism diagnostics from the scored panel, not the sealed confirmation. The data-driven version of this idea is Figure 1 of the article.

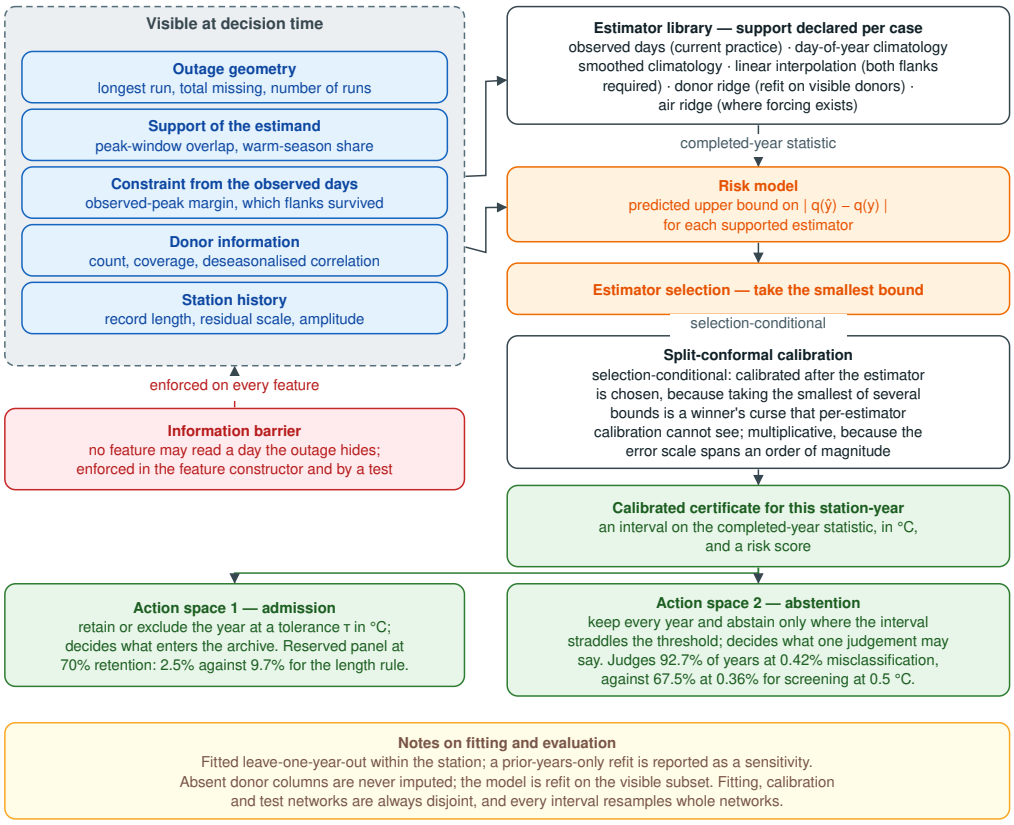


Figure S3. The screening system in detail. Left: the features visible at decision time, under an information barrier that forbids any feature from reading a day the gap hides. Centre: the estimator library with per-case support, the risk model, estimator selection, and the selection-conditional multiplicative conformal calibration. Bottom: the two action spaces the same calibrated object supports.

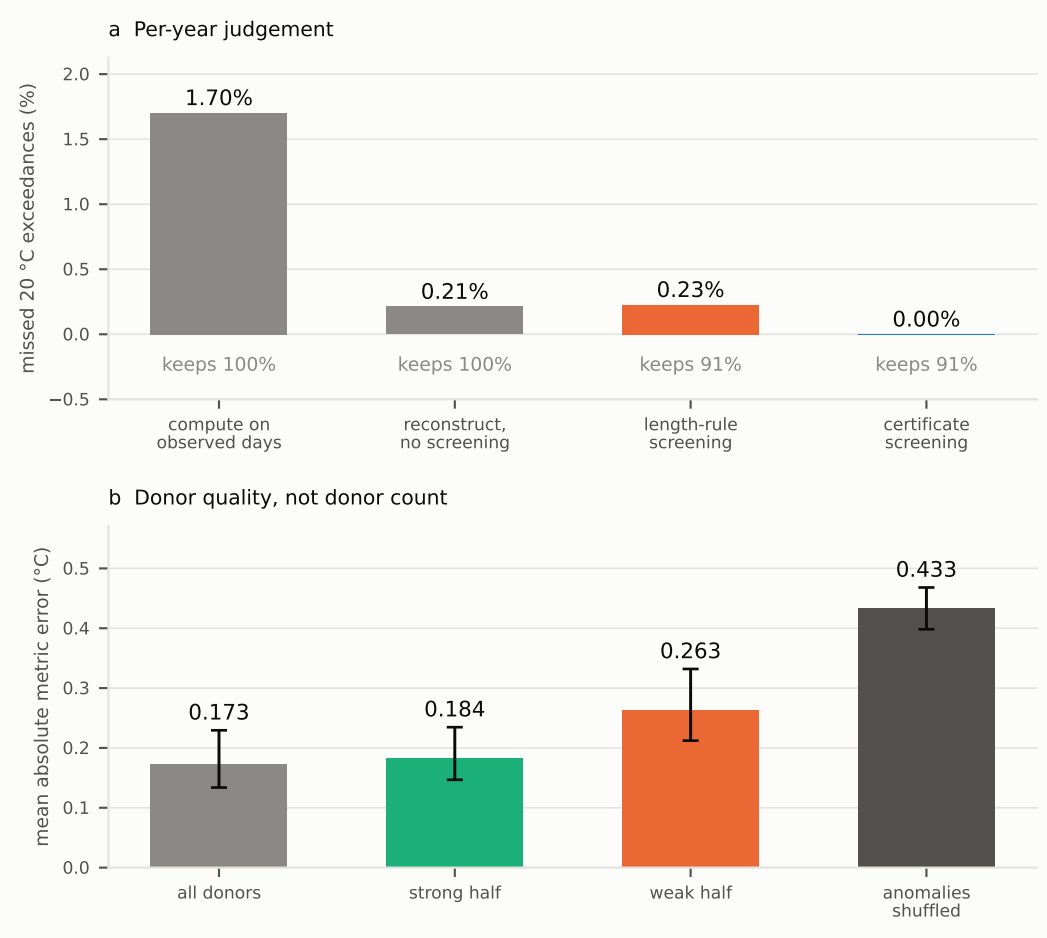


Figure S4. Multi-year trend error (a) and the donor controls (b). The panel averages eleven complete years per station, which is too short to resolve a decadal trend; the slope error is reported as a diagnostic of the reconstruction rather than as a trend estimate.

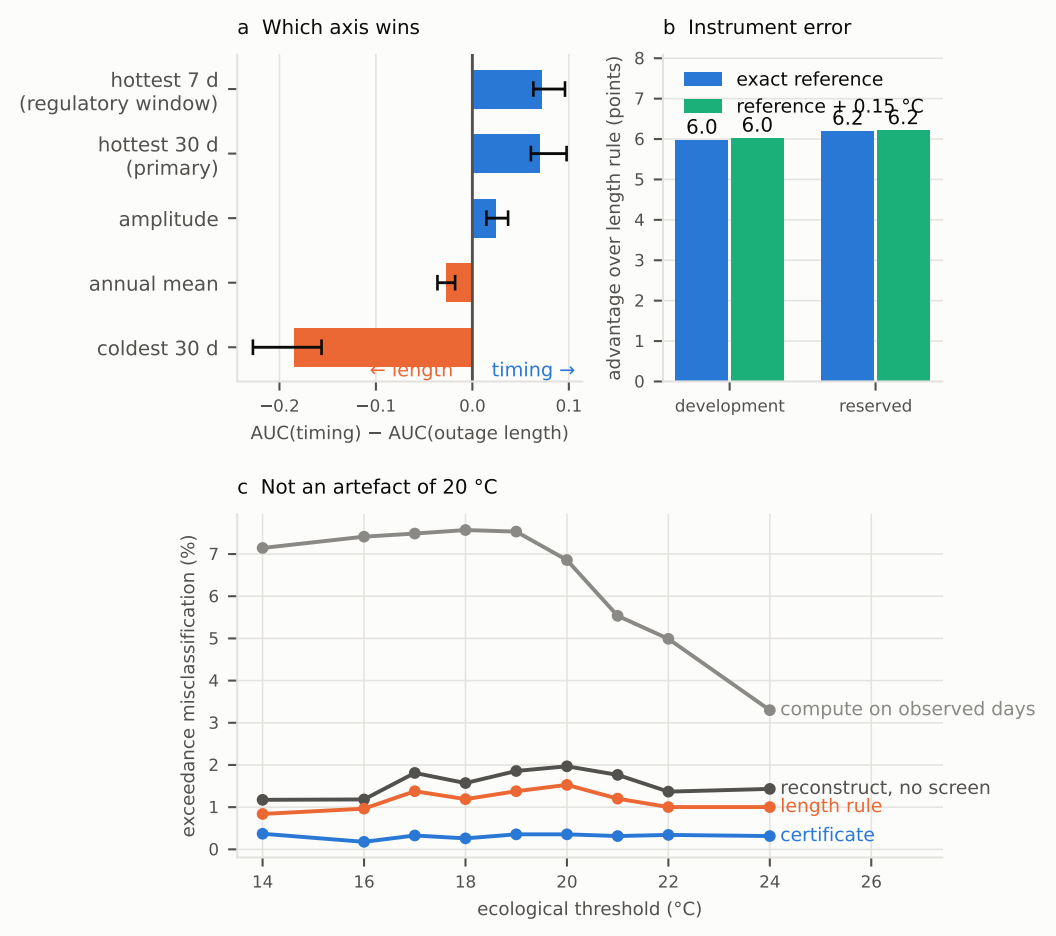


Figure S5. Development cases (n = 40 704, 110 networks; all intervals network-clustered 95%): AUC difference between the timing axis and the outage-length rule for each endpoint, in AUC units (a); the consequential-error advantage over the length rule in percentage points under a clean and a perturbed reference series, with the paired change interval (b); and the misclassification rate against the exceedance threshold in °C, per treatment (c). Panel (b) uses one reconstruction choice on both arms.

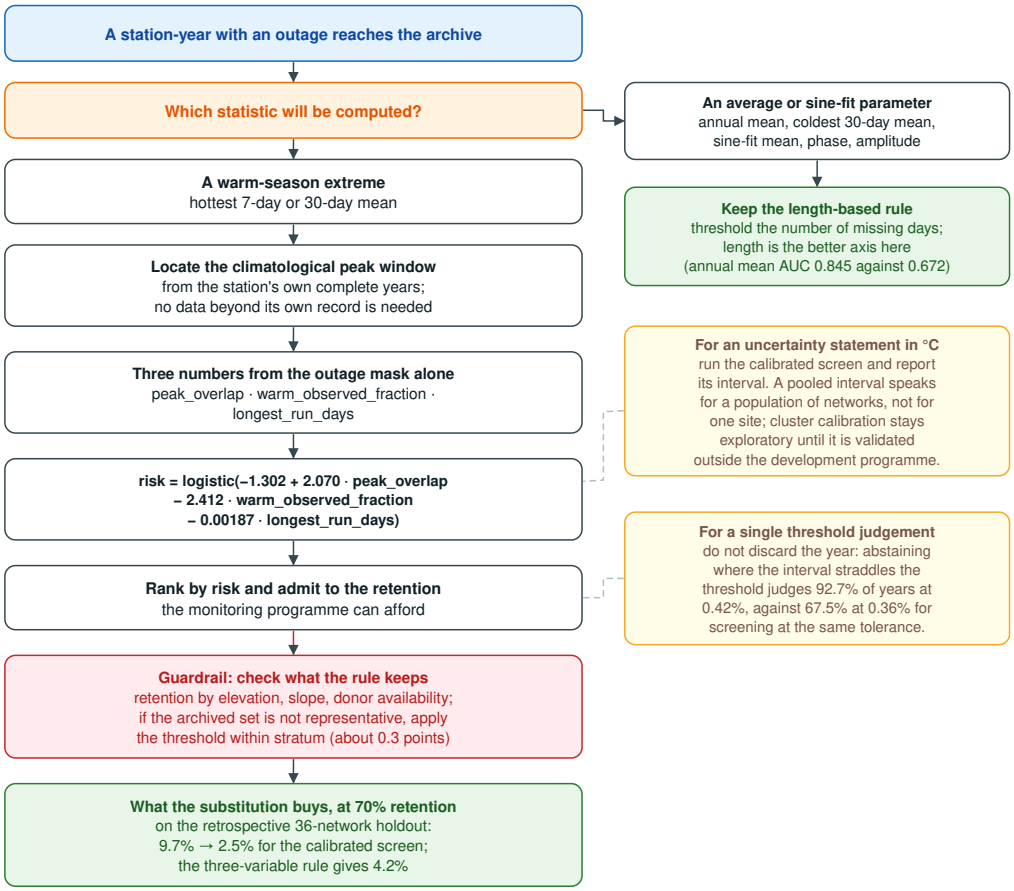


Figure S6. What an agency does with the result: which endpoint, where the peak window comes from, the three numbers of the deployable rule, the representativeness guardrail, and what the substitution buys on networks never used for fitting.

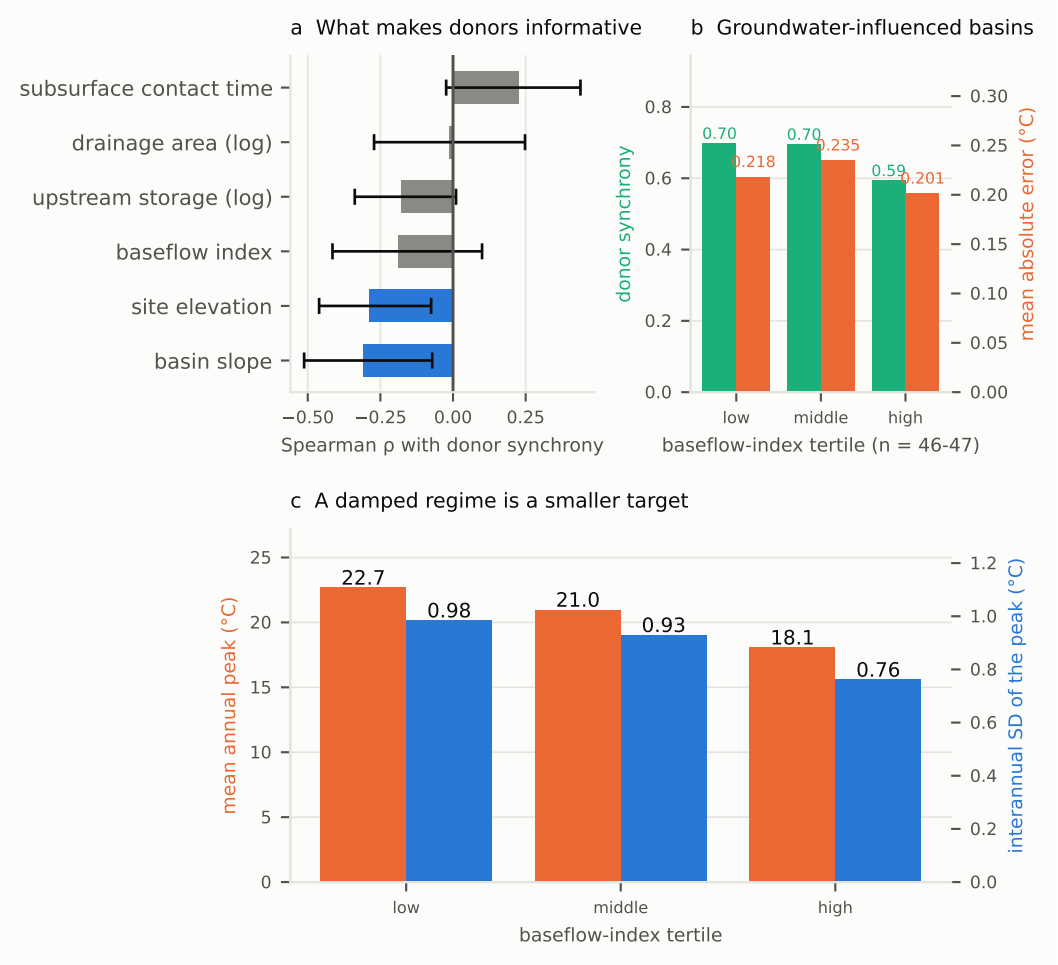


Figure S7. Catchment attribution on the United States subset, moved here from the article: 140 of 204 stations matched to GAGES-II across 57 networks. Donor synchrony against site elevation (a), completed-year absolute error (°C) against elevation (b), and the decomposition of the baseflow-index effect (c). Points are stations; bands are network-clustered 95% intervals. These are attribute associations on one country and are not a geology law.



Figure S8. Study design and data flow. Nine national providers are ingested and quality-checked into 146 river-only networks, 457 stations and 5114 complete station-years; 51 524 designed outages are placed on those complete years and split by network into a 110-network development set and a 36-network reserved panel, and the configuration is frozen. The reserved panel is a retrospective holdout with mixed prior access. The first sealed USGS and multi-agency transport panels fail the 2026-10-13 audits and are withdrawn; an independent wave-2 confirmation (49 networks; USGS, Hub'Eau, UK EA) meets the pre-declared gate under repaired QA. The same frozen configuration is applied, without truth, to 2422 real incomplete station-years, with admission requiring a finite annual metric. Source: draw.io source `figures/v17/drawio/f2_experiment_design.drawio`.